

Evaluation of Intelligent Systems

ARTIFICIAL INTELLIGENCE

Machines performing tasks that normally require human intelligence.

- Rule-Based Systems
- Search Algorithms (BFS, DFS)
- Logic & Reasoning
- Expert Systems

- Game Playing (Chess)
- Robotics
- Expert Systems
- Planning & Scheduling

MACHINE LEARNING

Machines learn from data and improve with experience.

- Linear Regression
- Decision Trees
- Support Vector Machines
- k-Means Clustering
- Naïve Bayes
- Reinforcement Learning

- Spam Detection
- Recommendation Systems
- Fraud Detection
- Sales Forecasting
- Customer Segmentation

DEEP LEARNING (DL)

Uses neural networks with many layers to learn complex patterns.

- Artificial Neural Networks
- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN)
- Long Short-Term Memory (LSTM)
- Transformers (Basic)

- Image Recognition
- Speech Recognition
- Machine Translation
- Medical Image Analysis
- Autonomous Vehicles

GENERATIVE AI

AI models that generate new content like text, images, audio, code.

- Generative Adversarial Networks (GANs)
- Variational Autoencoders (VAEs)
- Diffusion Models
- Transformers (Generative)

- Text Generation
- Image / Video Generation
- Code Generation
- Music & Audio Generation
- AI Art & Design

LARGE LANGUAGE MODELS (LLMs)

Advanced generative models trained on massive text data to understand and generate human language.

- Transformers (Advanced)
- Pre-training
- Fine-tuning
- Reinforcement Learning from Human Feedback (RLHF)

- Chatbots & Virtual Assistants
- Question Answering
- Summarization
- Translation
- Code Assistant

KEY ALGORITHMS

APPLICATIONS

AI TECHNOLOGIES COMPARISON

